

Newton–Raphson for the MLE

Where it appears in the course, and how to do it

Parametric Inference, B.Stat. 3rd Year — Lectures 7, 8, 10 and Homework 2 Q4

Contents

1	Where this appears in the course	1
2	Why the Cauchy needs it, and nothing else does	2
3	The two iterations	2
3.1	Fisher information for the Cauchy	3
3.2	Why you start at the sample median	3
4	Worked example	3
5	Code	4
6	When Newton–Raphson fails — and scoring does not	4
7	Beyond the Cauchy (not examinable, but worth seeing)	5
8	Related questions, with sources	6

1 Where this appears in the course

Narrower than it looks. Three mentions in the notes, and exactly **one** place where you are actually required to compute anything.

Where	What it says	Status
Lecture 8 (21 Aug), p.1	“Method of Scoring for computing MLE”: the ML equation, the Newton–Raphson iteration, and replacing the second derivative by its expectation (the Fisher information)	The only place the <i>method</i> is taught
Lecture 7 (sheet dated 10 Aug), p.3	Cauchy: “what is ML eq? We get a poly eq with no closed sol.”	Motivation only
Lecture 10 (28 Aug), p.1	“For any problem involving Cauchy distribution, starting with sample median is good”	Starting-value advice
Homework 2, Q4	Generate Cauchy(0) data for $n = 10, 15, 20$ and compute the MLE	The only place you must run it

It is never set as a # class exercise

All 34 of the professor’s “#” exercises are catalogued in `PI_Class_Exercises_Solved.pdf`. The Lecture 8 ones are: solve PS1 Q3 with σ^2 unknown; derive the Bayes estimate $\frac{p+X}{p+q+n}$; show $\max_{\theta} R(\theta, X/n) = \frac{1}{4n}$; complete the constant-risk minimax proof; and the improper-prior equalizer question. **None involves Newton–Raphson.**

So: theory in Lecture 8, practice in Homework 2 Q4, nothing else. The full solution to that

homework question, with code and output, is in `PI_Assignment_Solutions.pdf` §2.

2 Why the Cauchy needs it, and nothing else does

Every other family in the syllabus has an MLE you can write down: $\bar{X}$ for the Normal, Poisson and Exponential; X/n for the Binomial; $X_{(n)}$ for $U[0, \theta]$. The Cauchy does not.

With $f(x | \theta) = \frac{1}{\pi\{1 + (x - \theta)^2\}}$,

$$\ell(\theta) = - \sum_{i=1}^n \log \{1 + (x_i - \theta)^2\} + \text{const},$$

$$\ell'(\theta) = \sum_{i=1}^n \frac{2(x_i - \theta)}{1 + (x_i - \theta)^2}, \quad \ell''(\theta) = \sum_{i=1}^n \frac{2[(x_i - \theta)^2 - 1]}{[1 + (x_i - \theta)^2]^2}.$$

Why there is no closed form

Clear denominators in $\ell'(\theta) = 0$:

$$\sum_{i=1}^n 2(x_i - \theta) \prod_{j \neq i} \{1 + (x_j - \theta)^2\} = 0.$$

Each summand is a degree-1 factor times $n - 1$ quadratics, so this is a polynomial equation of degree $2n - 1$ in θ . For $n \geq 3$ there is no general solution by radicals, and worse, the equation can have **several roots** — so “the MLE” means the *global* maximiser, not just any stationary point. Both facts are why the course introduces a numerical method at all.

3 The two iterations

Newton–Raphson

Apply Newton’s root-finding method to $\ell'(\theta) = 0$:

$$\theta^{(m+1)} = \theta^{(m)} - \frac{\ell'(\theta^{(m)})}{\ell''(\theta^{(m)})}.$$

Converges *quadratically* near a root — typically 3–4 iterations to machine precision.

Method of scoring (Lecture 8)

Replace the observed second derivative by its expectation. Since $\mathbb{E}_\theta[\ell''(\theta)] = -nI(\theta)$,

$$\theta^{(m+1)} = \theta^{(m)} + \frac{\ell'(\theta^{(m)})}{nI(\theta^{(m)})}.$$

The professor’s stated reason for the substitution is that $I(\theta)$ is often free of θ — in particular **whenever θ is a location parameter** — which makes the iteration cheaper. But there is a second, more important reason, and it is the point of §6: *$-nI(\theta)$ is always negative, so every scoring step is guaranteed to point uphill.* Newton–Raphson carries no such guarantee.

3.1 Fisher information for the Cauchy

With $u = x - \theta$, the score is $\frac{\partial}{\partial \theta} \log f = \frac{2u}{1+u^2}$, so

$$I(\theta) = \mathbb{E} \left[\frac{4U^2}{(1+U^2)^2} \right] = \frac{4}{\pi} \int_{-\infty}^{\infty} \frac{u^2}{(1+u^2)^3} du.$$

Substituting $u = \tan t$ (so $du = \sec^2 t dt$ and $1+u^2 = \sec^2 t$),

$$\int_{-\infty}^{\infty} \frac{u^2}{(1+u^2)^3} du = \int_{-\pi/2}^{\pi/2} \frac{\tan^2 t}{\sec^4 t} dt = \int_{-\pi/2}^{\pi/2} \sin^2 t \cos^2 t dt = \frac{1}{4} \int_{-\pi/2}^{\pi/2} \sin^2(2t) dt = \frac{\pi}{8},$$

so

$$I(\theta) = \frac{4}{\pi} \cdot \frac{\pi}{8} = \frac{1}{2}$$

a *constant*, free of θ — exactly the location-parameter phenomenon Lecture 8 mentions. (Monte Carlo check with 4×10^6 draws: 0.4999.) The scoring iteration therefore collapses to something you can run on paper:

$$\theta^{(m+1)} = \theta^{(m)} + \frac{2}{n} \ell'(\theta^{(m)})$$

with no second derivative anywhere.

3.2 Why you start at the sample median

Lecture 10 says to start from $\tilde{X}_n$. The reason is that the obvious alternative is useless: the Cauchy is *stable*, so $\bar{X} \sim \text{Cauchy}(\theta, 1)$ for *every* n — averaging does not concentrate at all, and $\bar{X}$ is not even consistent. The sample median *is* consistent (Lecture 10), so it is a starting value that is already in the right neighbourhood.

4 Worked example

Take $n = 5$ with

$$x = (-2.1, -0.6, 0.3, 1.2, 4.8), \quad \text{median} = 0.300, \quad \bar{X} = 0.720.$$

The mean is already dragged off by the single observation 4.8 — a preview of why $\bar{X}$ is the wrong tool here. Start both iterations at $\theta^{(0)} = 0.300$.

Newton–Raphson $\theta^{(m+1)} = \theta^{(m)} - \ell' / \ell''$				
m	$\theta^{(m)}$	$\ell'(\theta^{(m)})$	$\ell''(\theta^{(m)})$	step
0	0.300000	-0.286530	-1.938398	-0.147818
1	0.152182	-0.003068	-1.881721	-0.001630
2	0.150552	-0.000001	-1.880643	-0.000001
3	0.150551	≈ 0	-1.880642	—

$$\hat{\theta} = 0.1506$$

Three iterations. Note the quadratic convergence: the step shrinks from 10^{-1} to 10^{-3} to 10^{-6} . A grid search over $[-6, 8]$ confirms the global maximiser is 0.15056.

Method of scoring $\theta^{(m+1)} = \theta^{(m)} + 0.4 \ell'(\theta^{(m)})$							
m	0	1	2	3	4	5	6
$\theta^{(m)}$	0.3000	0.1854	0.1590	0.1526	0.1511	0.1507	0.1506

Same answer, but convergence is *linear* — the error falls by a roughly constant factor each step rather than squaring. That is the trade: scoring is slower but needs no second derivative.

	sample mean	sample median	MLE
estimate of θ	0.720	0.300	0.151

5 Code

```
import numpy as np
rng = np.random.default_rng(2026)

def score(t, x): return np.sum(2*(x-t)/(1+(x-t)**2))
def hess (t, x): return np.sum(2*((x-t)**2 - 1)/(1+(x-t)**2)**2)

for n in (10, 15, 20): # Homework 2 Q4
    x = rng.standard_cauchy(n) # theta = 0
    t = np.median(x) # consistent starting value
    for _ in range(100):
        step = score(t, x)/hess(t, x)
        t -= step
        if abs(step) < 1e-12: break
    print(n, np.median(x), t, x.mean())
```

```
set.seed(2026)
for (n in c(10,15,20)) {
  x <- rcauchy(n)
  nll <- function(t) sum(log(1 + (x-t)^2))
  fit <- optimize(nll, interval = c(min(x), max(x)))
  cat(n, median(x), fit$minimum, mean(x), "\n")
}
```

Always verify against a grid or a built-in optimiser. The likelihood can be multimodal, and a single Newton run tells you nothing about whether you found the global maximum.

6 When Newton–Raphson fails — and scoring does not

This is the most instructive example in the document, and it justifies the Lecture 8 substitution far better than “it is cheaper”.

Take four observations in two well-separated clusters:

$$x = (-5, -4.5, 4.5, 5), \quad \text{median} = 0.$$

The log-likelihood is bimodal, with *three* stationary points:

θ	$\ell(\theta)$	$\ell''(\theta)$	type
−4.6216	−9.1205	< 0	global maximum
0.0000	−12.6289	+0.3125	local MINIMUM
+4.6216	−9.1205	< 0	global maximum

The sample median sits exactly on the local minimum

Newton–Raphson solves $\ell' = 0$. It does **not** care whether the root is a maximum or a minimum. Started at $\theta^{(0)} = 0$ we have $\ell'(0) = 0$ already, so the step is zero and the iteration *never moves* — it reports a local minimum as the answer.

Worse, this is not a knife-edge accident. Starting at $\theta^{(0)} = 4.0$, well inside the right-hand mode:

$$4.000 \rightarrow 5.482 \rightarrow 3.293 \rightarrow 0.330 \rightarrow 0.002 \rightarrow 0.$$

It overshoots to 5.48, comes back to 3.29 where ℓ'' has flipped sign to *positive*, and from there gets dragged into the minimum. Only a start at 4.5 — already very close — converges correctly to 4.6216.

Scoring succeeds, and here is why

The scoring update $\theta^{(m+1)} = \theta^{(m)} + \frac{2}{n}\ell'(\theta^{(m)})$ from $\theta^{(0)} = 0.1$ converges to 4.6216: the global maximum.

The reason is structural. Scoring divides by $-nI(\theta)$, which is **always negative**, so the step is always a positive multiple of ℓ' — i.e. always in the direction of *increasing* likelihood. It is a gradient ascent with a principled step size and **cannot be attracted to a minimum**. Newton–Raphson divides by ℓ'' , whose sign flips wherever the likelihood is locally convex, and there the step points the wrong way.

So the Lecture 8 substitution buys more than cheapness: it buys an **ascent guarantee**.

Practical rules

1. Start from the sample median (Lecture 10) — but treat it as a suggestion, not a guarantee. Check $\ell'' < 0$ there.
2. Prefer **scoring** when the likelihood may be multimodal; prefer **NR** for speed once you are already near the maximum. Many implementations do scoring first, then switch.
3. **Always confirm the answer is a global maximum** by evaluating ℓ on a grid, or by restarting from several points.
4. If NR produces a θ with $\ell''(\theta) > 0$, you have found a minimum. Discard it.

7 Beyond the Cauchy (not examinable, but worth seeing)

The Cauchy is the course’s only instance, which can leave the impression that Newton–Raphson is a Cauchy trick. It is not. The Gamma with *both* parameters unknown — whose exponential-family form you derived in `PI_Exponential_Families_Crash_Course.pdf` — is another natural case.

For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Gamma}(\alpha, \beta)$ (shape, scale),

$$\ell(\alpha, \beta) = (\alpha - 1) \sum_i \log x_i - \frac{1}{\beta} \sum_i x_i - n \log \Gamma(\alpha) - n\alpha \log \beta.$$

Setting $\partial\ell/\partial\beta = 0$ gives $\hat{\beta} = \bar{X}/\alpha$; substituting into $\partial\ell/\partial\alpha = 0$ leaves a *one-dimensional* equation,

$$\log \alpha - \psi(\alpha) = s, \quad s := \log \bar{X} - \frac{1}{n} \sum_i \log x_i = \log \frac{\text{AM}}{\text{GM}} > 0,$$

where ψ is the digamma function. No closed form — so, Newton–Raphson, with

$$g(\alpha) = \log \alpha - \psi(\alpha) - s, \quad g'(\alpha) = \frac{1}{\alpha} - \psi'(\alpha).$$

Worked instance. $x = (1.2, 3.4, 0.8, 5.1, 2.6, 1.9, 4.3, 2.2)$, $n = 8$: $\bar{X} = 2.6875$, $\frac{1}{n} \sum \log x_i = 0.8321$, so $s = 0.15653$. The standard starting approximation $\alpha_0 = \frac{3 - s + \sqrt{(s - 3)^2 + 24s}}{12s}$ gives 3.34579, and then

m	$\alpha^{(m)}$	$g(\alpha^{(m)})$
0	3.345793	$+2.89 \times 10^{-4}$
1	3.351692	$+5.31 \times 10^{-7}$
2	3.351703	$+1.80 \times 10^{-12}$

$$\hat{\alpha} = 3.3517, \quad \hat{\beta} = \bar{X}/\hat{\alpha} = 0.8018.$$

Cross-checked against a bracketing root-finder on $[0.1, 50]$: identical to six decimals.

The moral

The MLE has no closed form here for the same underlying reason as in the Cauchy: the likelihood equation involves a transcendental function of the parameter. Note the connection to the exponential-family derivation — since $T_1(x) = \log x$ for the Gamma, the natural estimating equation is about $\mathbb{E}[\log X] = \psi(\alpha) + \log \beta$, and inverting ψ is exactly what has no closed form.

8 Related questions, with sources

- Q1.** [Homework 2 Q4 — solved in [PI_Assignment_Solutions.pdf §2](#)] Generate $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Cauchy}(0)$ for $n = 10, 15, 20$, compute the MLE, and compare with $\bar{X}$.
- Q2.** [Lecture 8 — worked in class] Write down the method-of-scoring update for the Cauchy location parameter and explain why $I(\theta)$ does not involve θ .
- Q3.** [Lecture 7 — stated in class, not set] Show the Cauchy likelihood equation is a polynomial of degree $2n - 1$, and deduce that the MLE need not be unique.
- Q4.** [Lecture 10 — stated in class] Why start the iteration from the sample median rather than the sample mean? (*Because $\bar{X} \sim \text{Cauchy}(\theta, 1)$ for every n , so it is not even consistent, while the median is.*)
- Q5.** [Class Exercise 10.2 — solved in [PI_Class_Exercises_Solved.pdf §10](#)] Prove consistency of the MLE in general, and explain why $U[0, \theta]$ is consistent despite failing the regularity conditions.
- Q6.** [New — §6 above] For $x = (-5, -4.5, 4.5, 5)$, find all stationary points of the Cauchy log-likelihood, classify each, and show that Newton–Raphson started at the sample median fails while the method of scoring succeeds.

What a written exam could realistically ask

Not “perform five iterations by hand” — that is a lab task. Far more likely, and worth roughly 8–10 marks:

Derive the likelihood equation for the Cauchy location parameter; show it has no closed-form solution; describe the Newton–Raphson and method-of-scoring iterations; compute $I(\theta)$ and hence give the scoring update explicitly; and state, with reason, a suitable starting value.

Every ingredient of that answer is in §2–§3 of this document. That is also why Newton–Raphson does not appear in `PI_Mock_Exam.pdf`.

Companions: `PI_Assignment_Solutions.pdf` (HW2 Q4), `PI_Class_Exercises_Solved.pdf` §10, `PI_Exponential_Families_Crash_Course.pdf`, `PI_Exam_Revision.pdf` §12–§13.